

# Deepak Mallampati

## APPLIED AI / LLM ENGINEER

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Kent, OH | US-based, F-1 OPT, open to sponsorship

### PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years owning AI quality end-to-end - defining what "good" means, measuring it, and shipping what works. Built a rigorous LLM evaluation framework (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards, raised task-success rates **27%** through prompt and orchestration iteration, and reduced hallucinations **40%** at **92% precision**. Comfortable embedding in product teams across search, chat, and documents with experiment-driven rigor.

### CORE COMPETENCIES & SKILLS

Rigorous Evaluation • Prompt & Orchestration Design • Rapid Experimentation / A/B • AI Quality Ownership • LLM Product Features • RAG (Search / Docs) • Python • Researcher Collaboration

**Generative AI & LLMs:** GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety | Hugging Face Transformers

**Machine Learning & NLP:** PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore

**Vector Search & Retrieval:** FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | Semantic Caching

**MLOps & Cloud:** AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, Weights & Biases | Model Monitoring & Observability

**Data Engineering:** PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance

**Security & Compliance:** HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Zero-Trust

**Languages & Tools:** Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

### PROFESSIONAL EXPERIENCE

#### Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) to enable compliant analytics and model development with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark iterations and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

## Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

## Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed and trained supervised **regression and classification models** on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Engineered features and executed end-to-end data preprocessing - cleaning, normalization, and transformation of raw sensor and operational datasets - to produce model-ready training pipelines.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific financial datasets, achieving **89% F1-score** on custom NER tasks.

## Suvidha Foundation

Jan 2022 - Mar 2022

ML Intern

India

- Designed an end-to-end **video summarization and highlight-generation pipeline**, integrating speech-to-text, transformer-based abstractive summarization, and timestamp-aligned clip extraction.
- Deployed the pipeline across **5,000+ videos**, automating highlight selection and reducing manual editing effort by **70%**.

## PROJECTS

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### MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

### career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - compressing hours of repetitive job-hunting into an automated workflow.

### Privacy-Preserving Clinical ML Pipeline

A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy, and quantifying the privacy-utility trade-off of each technique.

### Story Director LLM

A fine-tuned language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

RESEARCH

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity  $k=3$ , l-diversity  $l=2$ , Laplace differential privacy across four  $\epsilon$  levels), reducing record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset for controllable long-form generation; authored a peer-review-ready IEEE-format conference paper documenting methodology and results.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0

WORK AUTHORIZATION

Work Authorization - F-1 OPT, authorized to work in the United States; future sponsorship required